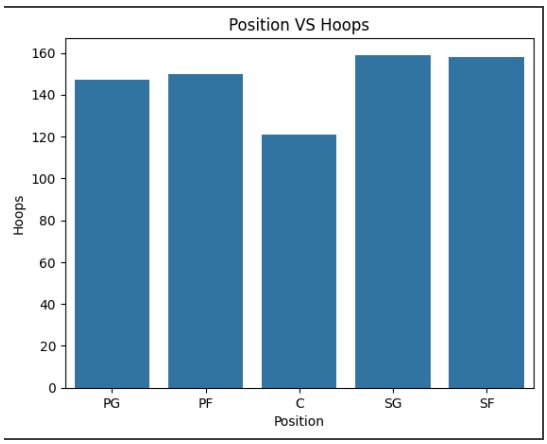
Results:

1. Data Extraction: Scraped NBA player statistics for the 2024 season from Basketball-Reference. The data included various statistics like points, percentages, rebounds, assists, blocks, and steals. You loaded this data into a pandas DataFrame and saved it to a CSV file.
2. Data Cleaning: Identified that the 'Name' column had all missing values and did not make sense so I dropped it. Also converted the relevant statistics columns to a numeric data type, handling potential errors by coercing them to NaN.
3. Outlier Removal and Descriptive Statistics: Identified an issue with the mode being 0 for numerical columns. To address this and clean the data further, and implemented a function to remove outliers using the Interquartile Range (IQR) method for the numerical columns. After removing outliers, I recalculated the mean, median, and mode, which showed more meaningful values for the mode.
4. Visualizations and Insights: I generated several plots to visualize the data and explore relationships:
    - A **count plot** showed the distribution of players across different positions.
    - A **stacked bar** chart visualized the sum of various statistics by position, highlighting the total contribution of each position in different categories.
    - A **box plot** of Assists by Position revealed the distribution and spread of assists, clearly showing that Point Guards have significantly higher assist numbers.
    - **Histograms** showed the distribution of Total Points and Field Goal Percentage, indicating a skewed distribution for points and a potentially multimodal distribution for field goal percentage.
    - **Scatter plots** of Assists vs. Steals and Points vs. Assists showed positive correlations between these pairs of statistics.

- A **pair plot** provided a quick overview of the relationships and distributions among scoring and shooting efficiency metrics.
- A **heatmap** visualized the correlation matrix of numerical statistics, highlighting strong positive correlations between points and various other statistics like assists, rebounds, blocks, and steals.

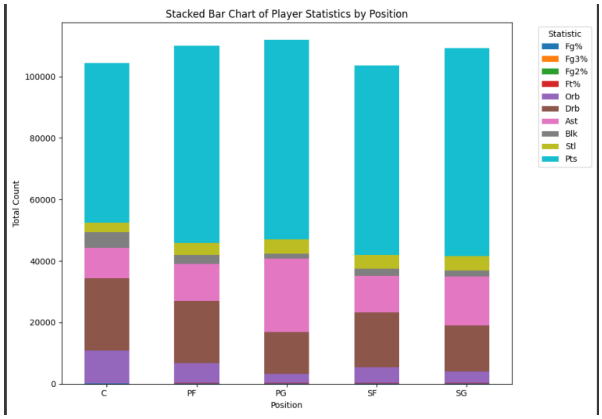
Here are some Insights we can derive using Visualizations we've generated

1. **Insight on Player Position Distribution:**



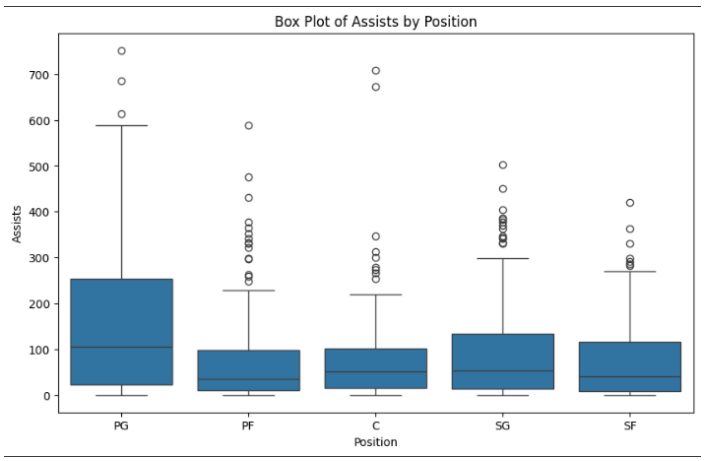
- Explanation: The count plot shows the number of players at each position (PG, PF, C, SG, SF). This plot helps us understand the distribution of players across different positions in the dataset. From the plot, we can see that there are roughly similar numbers of players in each position, with a slight variation.

2. Insight on Player Statistics Distribution by Position:



- Explanation: The stacked bar chart visualizes the sum of various statistics ('Fg%', 'Fg3%', 'Fg2%', 'Ft%', 'Orb', 'Drb', 'Ast', 'Blk', 'Stl', 'Pts') for each position. This plot allows us to compare the total contribution of each position across different statistical categories. For example, we can see which positions tend to have higher total points, assists, or rebounds. It appears that positions like C and PF have higher total rebounds (Orb and Drb), while PG and SG have higher total assists (Ast).

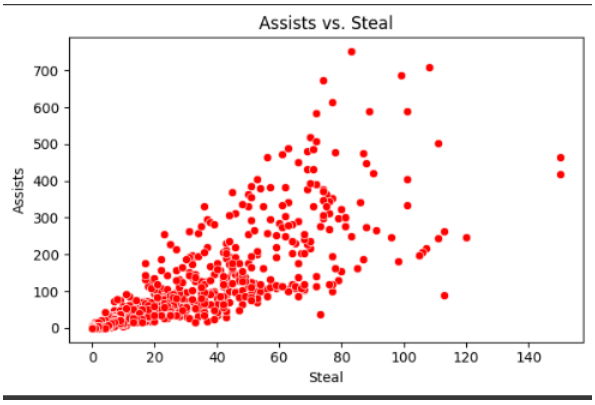
3. Insight on the Distribution and Spread of Assists by Position:



- Explanation: The box plot of Assists by Position shows the distribution, median, quartiles, and potential outliers for the 'Ast' statistic for each player position. We can see that Point Guards (PG) have a significantly higher median and a wider spread of assists compared to other positions, which is expected given their

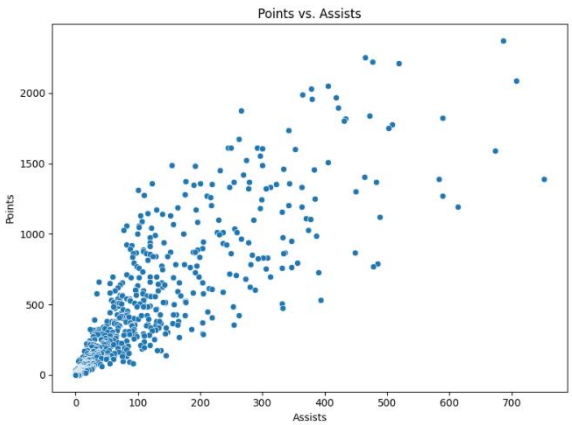
primary role in facilitating the offense. The plot also reveals several outliers for PG, indicating some players at this position have exceptionally high assist numbers.

4. **Insight on the Relationship between Assists and Steals:**



- Explanation: The scatter plot of Assists vs. Steals shows the relationship between these two statistics. There appears to be a positive correlation, meaning players with a higher number of assists also tend to have a higher number of steals, although there is a good amount of spread in the data. This could indicate that players who are more involved in the game flow on both ends of the court tend to accumulate both assists and steals.

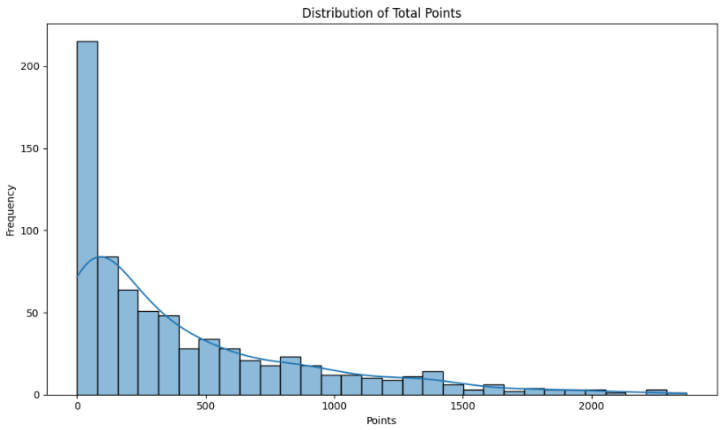
5. **Insight on the Relationship between Points and Assists:**



- Explanation: The scatter plot of Points vs. Assists shows the relationship between these two statistics. There is a clear positive correlation, indicating that players who score more points also

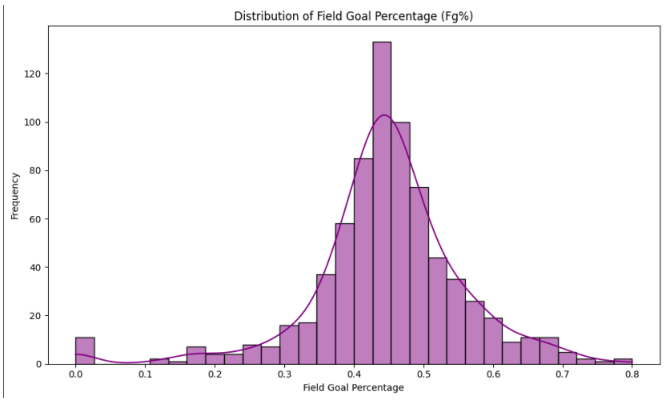
tend to have more assists. This is also expected, as primary ball-handlers and offensive threats often contribute significantly to both scoring and setting up teammates.

6. **Insight on the Distribution of Total Points:**



Explanation: This histogram shows the frequency distribution of the 'Pts' (Total Points) statistic across all players. We can see that the distribution is heavily skewed to the right, meaning a large number of players scored relatively few points, while a smaller number of players scored very high points. This is typical in sports statistics, where there are a few high-performing individuals and many players with more modest contributions.

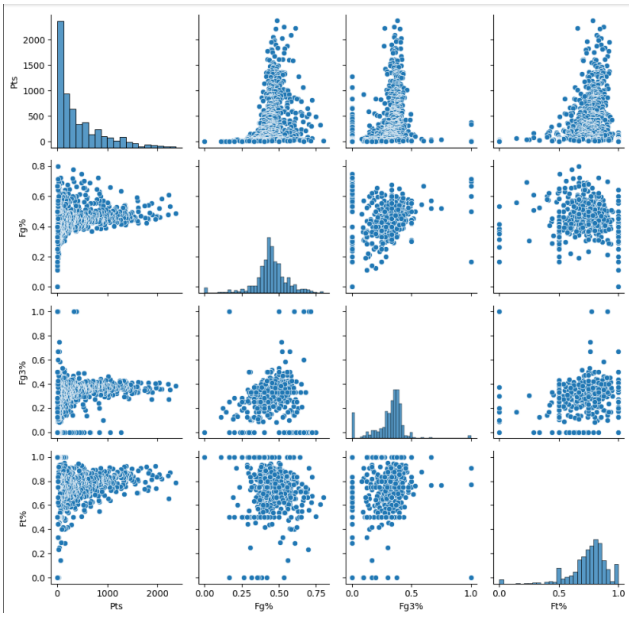
7. **Insight on the Distribution of Field Goal Percentage (Fg%):**



Explanation: This histogram shows the frequency distribution of 'Fg%' (Field Goal Percentage). The distribution appears somewhat bimodal or at least has a peak around 0.4 to 0.5, with smaller peaks or shoulders elsewhere. This suggests that there might be

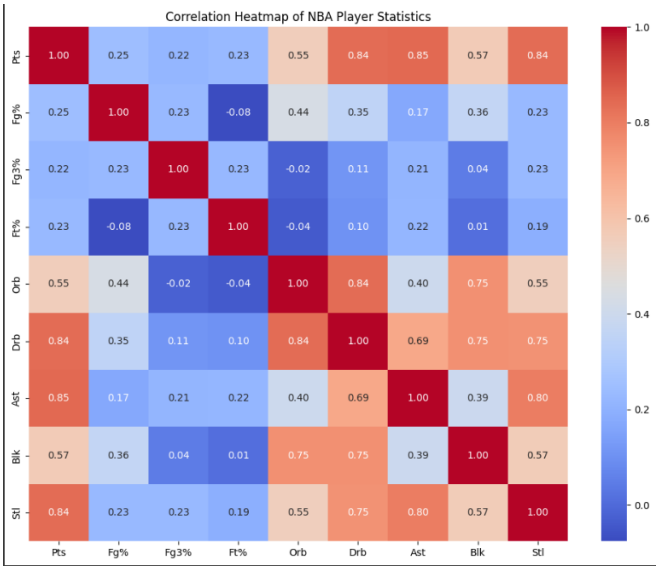
different groups of players with distinct shooting efficiencies, or that the distribution isn't a simple normal distribution.

8. **Insight on the Relationships between Scoring and Shooting Efficiency Metrics:**



Explanation: The pair plot displays scatter plots for all pairs of the selected numerical columns ('Pts', 'Fg%', 'Fg3%', 'Ft%') and histograms for each individual column along the diagonal. This plot is useful for quickly visualizing the relationships between multiple variables. We can observe the previously mentioned positive correlation between 'Pts' and 'Fg%', 'Pts' and 'Fg3%', and 'Pts' and 'Ft%'. It also allows us to see the distributions of each of these shooting percentages individually and how they relate to each other. For example, there appears to be a positive correlation between 'Fg%' and 'Fg2%', which is expected as 2-point shots are a component of the overall field goals.

9. Insight on the Correlation between Various Numerical Statistics:



Explanation: The heatmap visualizes the correlation matrix for the selected numerical columns ('Pts', 'Fg%', 'Fg3%', 'Ft%', 'Orb', 'Drb', 'Ast', 'Blk', 'Stl'). The color intensity and the numerical values in each cell indicate the strength and direction of the correlation between two variables. Red colors generally indicate positive correlation, while blue colors indicate negative correlation. The diagonal is always 1 (perfect positive correlation of a variable with itself). We can see strong positive correlations between 'Pts' and 'Ast', 'Pts' and 'Drb', 'Pts' and 'Stl', and 'Pts' and 'Blk'. This suggests that players who score more tend to contribute significantly in other statistical categories as well. We can also observe correlations between other pairs of statistics, such as a strong positive correlation between 'Orb' and 'Drb' (offensive and defensive rebounds) and between 'Ast' and 'Stl'.

RESULT

The analysis of the 2024 NBA player statistics data, through cleaning, descriptive statistics, and visualization, provides valuable insights into player performance and characteristics by position.

The data cleaning steps, including handling missing values and removing outliers, were crucial for ensuring the accuracy of the subsequent analysis. The descriptive statistics (mean, median, mode) for the cleaned data offer a better understanding of the typical values for various player statistics.

The visualizations effectively illustrate key trends and relationships:

- Player distribution across positions is **relatively balanced**.
- Different positions contribute differently to overall team statistics, with guards excelling in assists and forwards/centers in rebounds.
- **Point Guards are the primary playmakers**, as evidenced by their high assist numbers and the spread of the data in the box plot.
- There are positive relationships between scoring, assists, and defensive statistics like steals and blocks, suggesting that high-performing players often contribute across multiple facets of the game.
- The distribution of **points is heavily skewed**, indicating a performance hierarchy among players.

I scraped, cleaned, and visualized NBA stats. The plots showed player distribution by position, how stats vary by position, relationships between stats like assists and points, and identified outliers. Descriptive stats like mean and median were calculated.

Overall, this analysis provides a good understanding of the dynamics of NBA player statistics and how they vary by position.